Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:48 PM

Agenda

* Central Limit Theorem

** Assignments

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:08 PM

Normal Distribution

(1) Empirical Rule : $\mu + \sigma$

(1) $\mu - \sigma <= x <= \mu + \sigma$ → 68 %.

(2) $\mu - 2\sigma <= x <= \mu + 2\sigma$ → 95 %.

(3) $\mu - 3\sigma <= x <= \mu + 3\sigma$ → 99.7 %.

(2) z - Score

z-score table

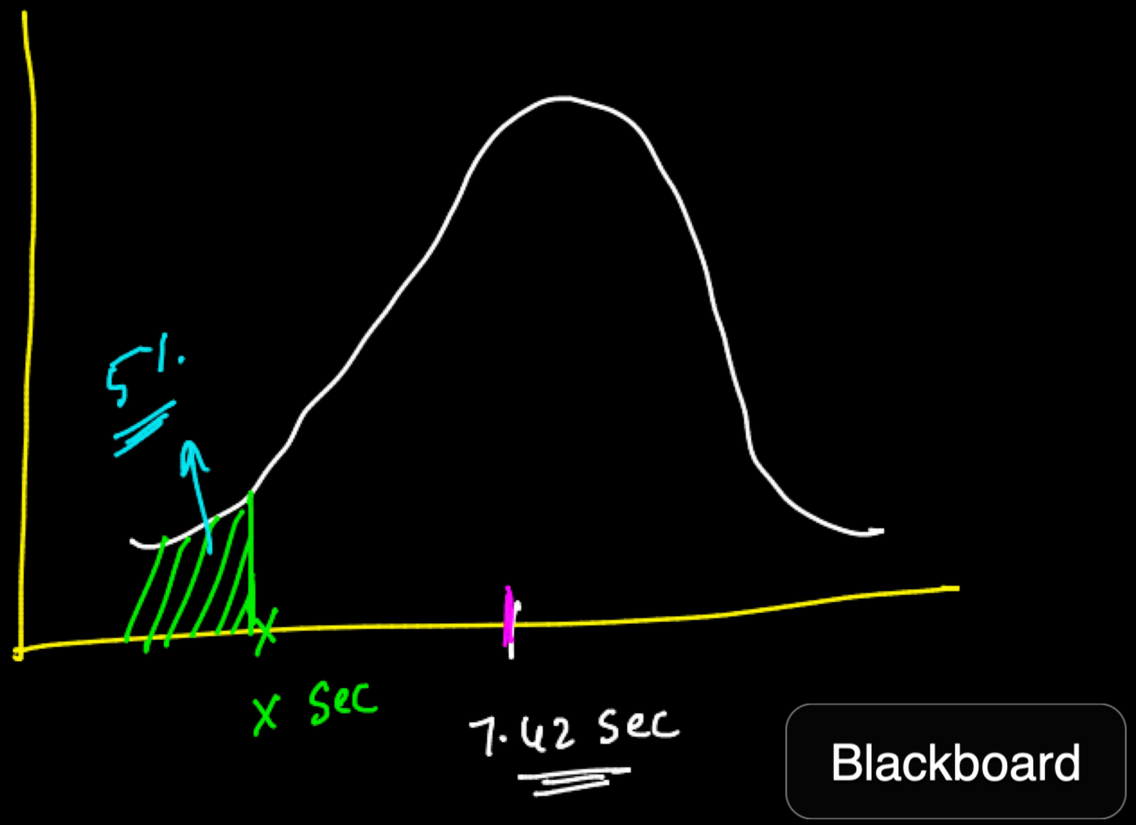
scipy

$$x = \mu + (\pm z) * \sigma$$

$$z = \frac{x - \mu}{\sigma}$$

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:19 PM

Quiz Skaters take a mean of 7.42 sec and Std-dev of 0.34 sec for 500 meters. What should his speed be such that he is faster than 95% of his competitors.



(1) what fraction of people slower than him 50%.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{500}{6.86}$$

Blackboard

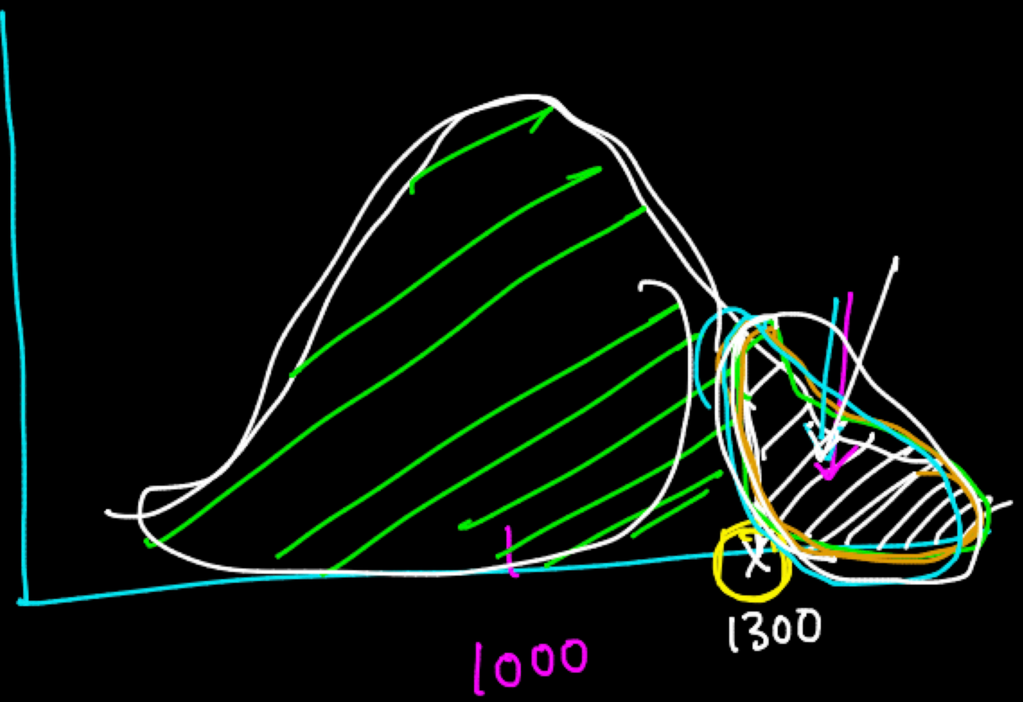
Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:08 PM

Quiz

⇒ retail outlet sells around 1000 tooth pastes a week.

std-dev = 200

on-hand inventory is 1300, What is need for replenishment with in a week ?



1 − norm.cdf(z)

0 → 1

Big , D-Mart ,

1st week

1000

weekly







$$1300 = 1000 + z * 200$$

$$z = \frac{1300 - 1000}{200} = \frac{300}{200} = 1.5$$

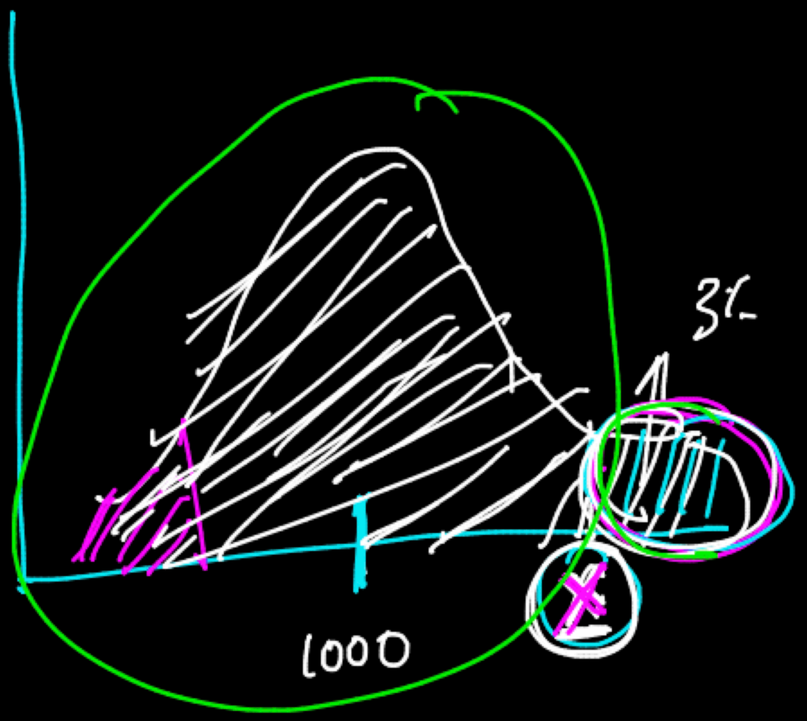
norm.cdf(z)

A new page has been created.



Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:09 PM

=> we want atmost (3%) of time we allow to be out of stock then what should be the on-hand inventory.



$\sigma = 200$

norm..

cdf

ppf (1 − 0.03)

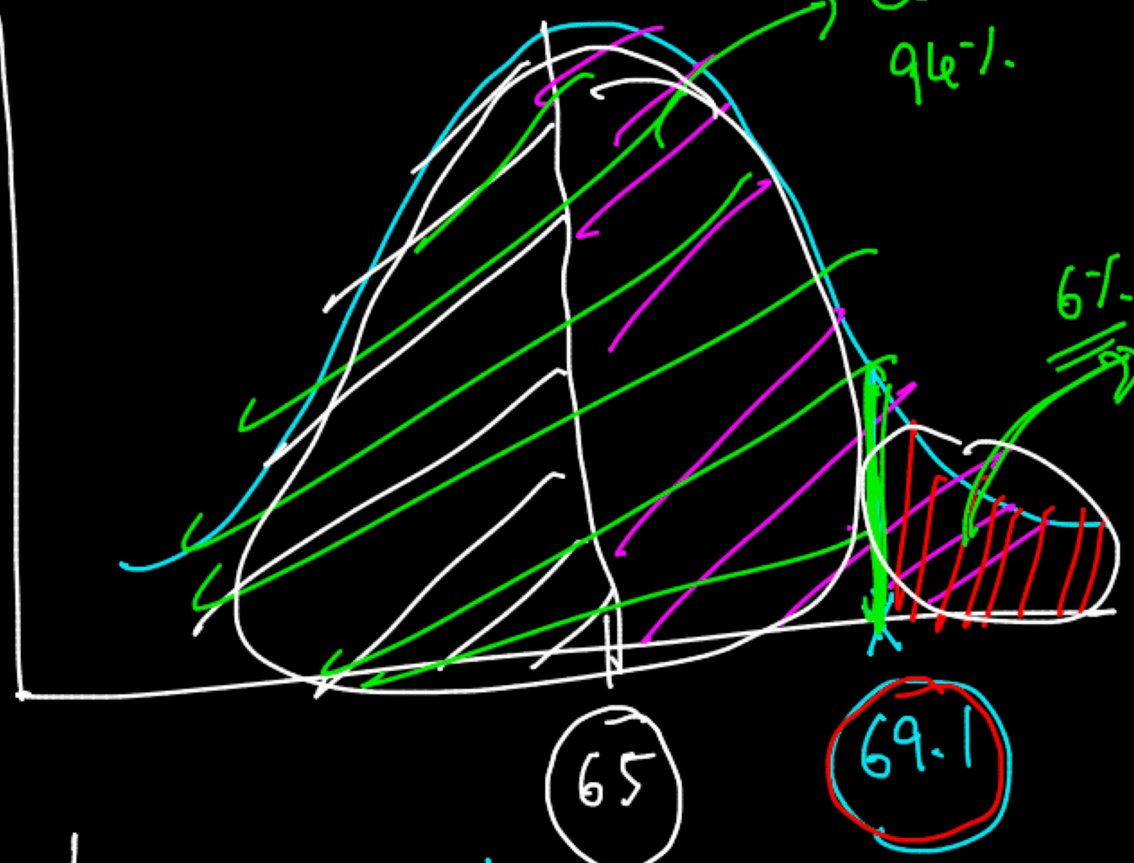
$x = 1000 + 1.88 * 200$

cdf(z)

1 − 0

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:47 PM

(1) =>



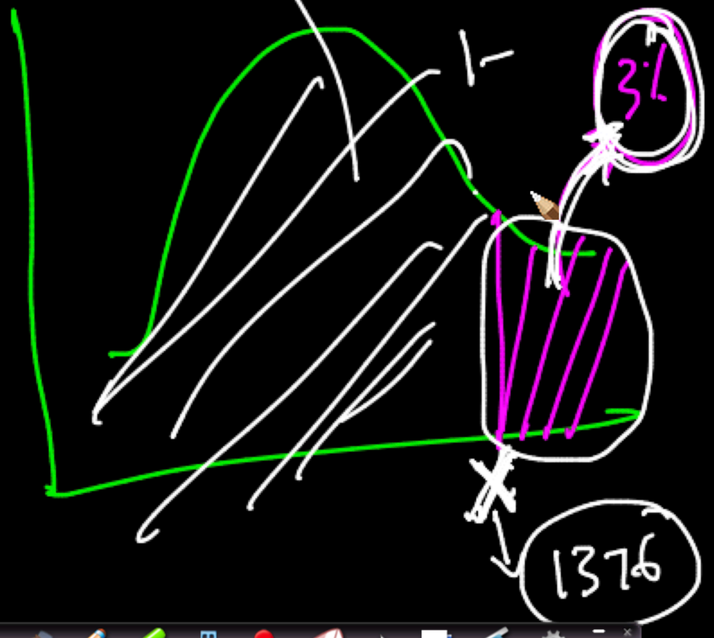
$\mu = 65$

$\sigma = 2.5$

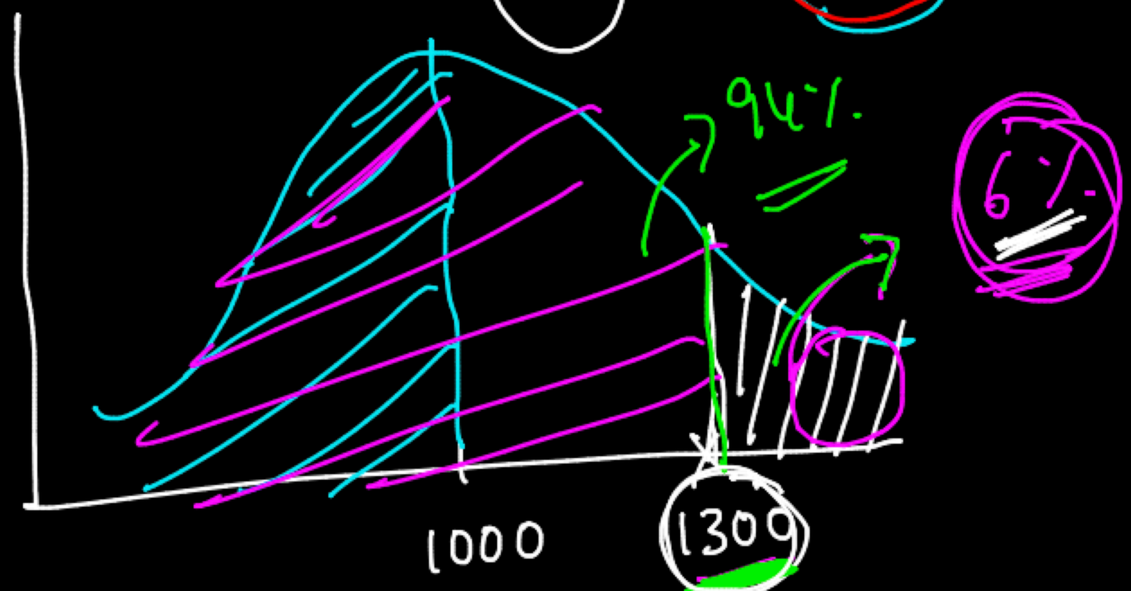


1300

1-



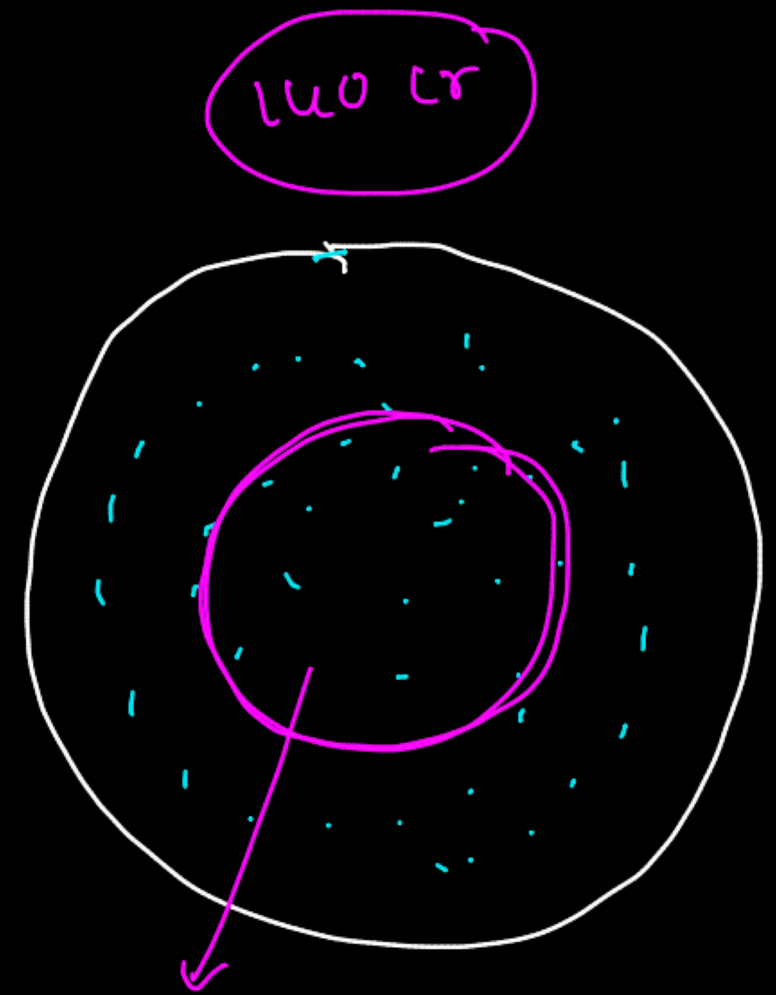
(2)



Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:50 PM

Population

India



Sample

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 9:55 PM

=> From samples, what can be inter about population

1000, 10000 (or) 1L

(1) Central limit theorem ✓

(2) Confidence Interval [next class]

10 / 10

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:32 PM

10000 → population

$\mu = 66.37$ Population
$\sigma = 3.84$

(1) randomly (5) persons height.

avg height

around 66

(2) randomly (5) persons

around 66 but diff than previous

5

3(d)4 < 66

3(d)4 > 66
≡

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:20 PM

Mean of sample mean ≈ population mean

1

$\sigma = 3.84$

Std deve of sample = 1.73

10:21 → 10:26

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:35 PM

if 'n' denotes the sample size

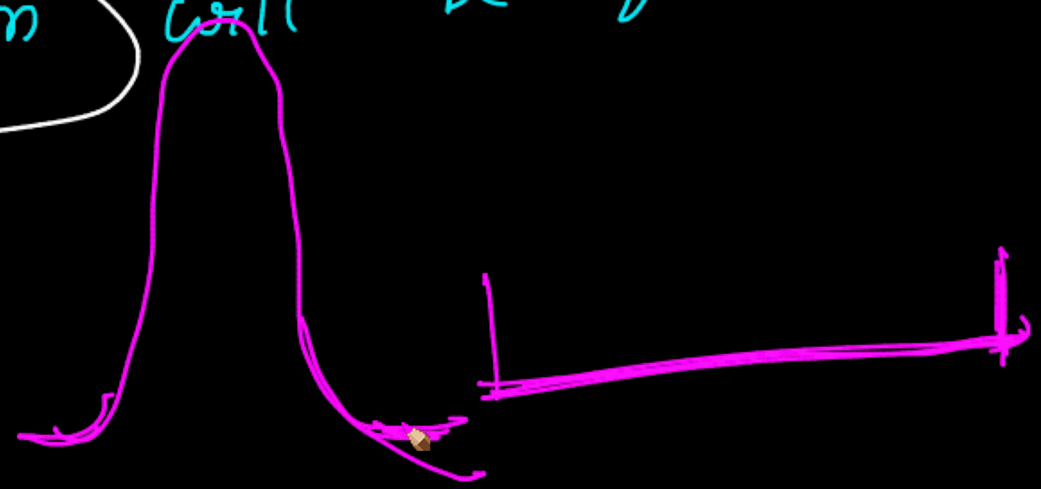
$$\left\{ \begin{array}{l} n = 5 \\ \quad 20 \\ \quad 100 \end{array} \right\} \Rightarrow$$

mean of sample mean $\simeq$ population

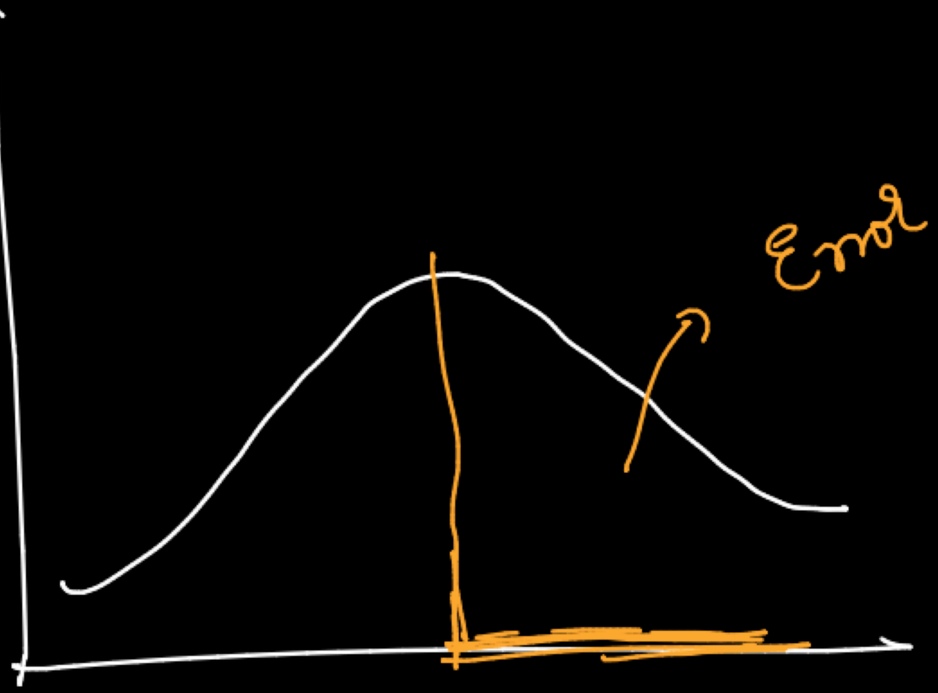
(6) $\Rightarrow$ population std dev

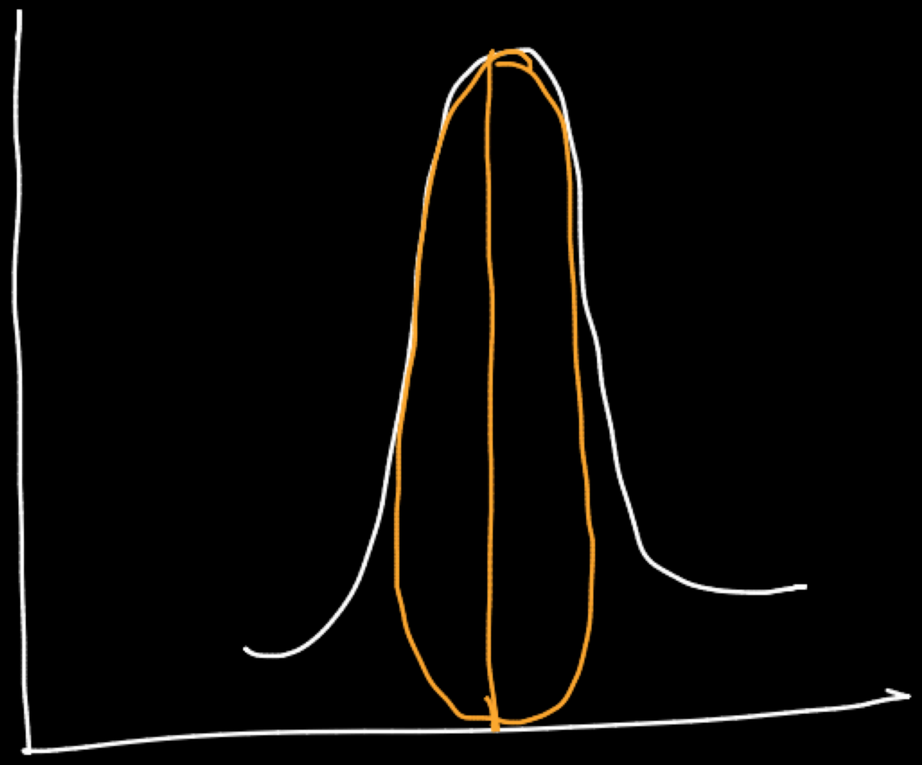
then std dev of sample mean will be equal to

Standard Error $= \frac{6}{\sqrt{n}}$



Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:39 PM





- mean of Sample means = population mean
- standard Error = $\frac{\sigma}{\sqrt{n}}$

14 / 15

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:53 PM

μ ⇒ population

Sample mean ← $\bar{x} = \dfrac{x_1 + x_2 + \cdots + x_n}{n}$

then $\bar{x}$ follows gaussian distribution with

$E(\bar{x}) = \mu$

Std devo of $\bar{x} = \dfrac{\sigma}{\sqrt{n}}$

mathematically

$\bar{x} \sim N\left(\mu, \dfrac{\sigma}{\sqrt{n}}\right)$

size

(1) $n > 30$ large

(2) σ ✓ finite

(3)

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:47 PM

⇒ Sampling distribution of the mean will be always normal distributed, as long as sample size is large enough. Regardless of whether the population has a normal (or) other distribution.

16 / 17

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 10:55 PM

⇒ Systolic blood pressure of a group of people is known to have an avg of 122 mm Hg and std dev. of 10 mm Hg. Calculate the probability that the avg blood pressure of 16 people will be greater than 125 mm Hg.

$\mu = 122$ mm Hg

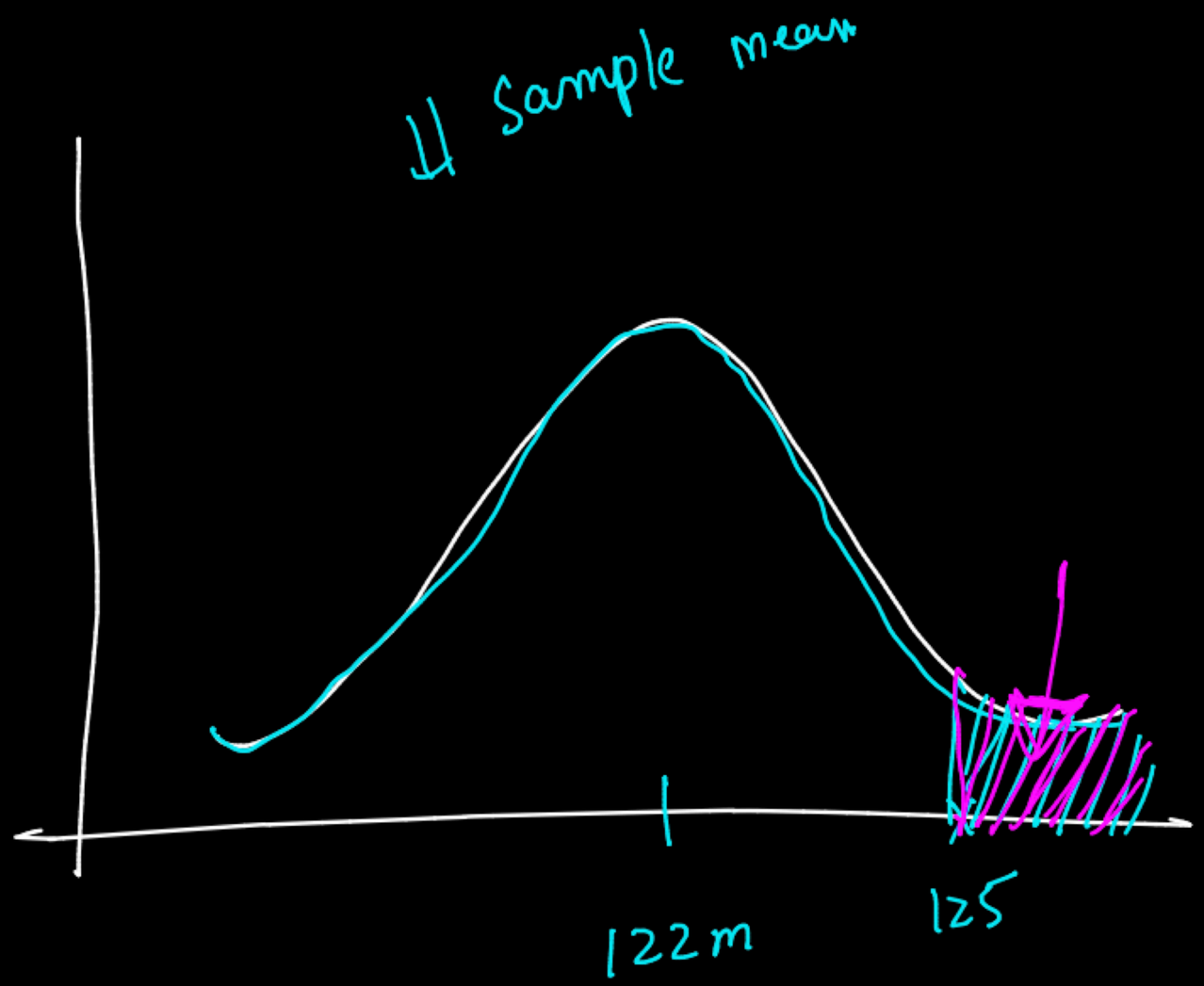
$\sigma = 10$ mm Hg

$n = 16$

$\bar{x} = 122$ mm Hg

Std-dev of sample mean $= \frac{\sigma}{\sqrt{n}} = \frac{10}{4} = 2.5$ m

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Mon 10 Jul 10:56 PM



Std devo

122

2.5

$$z = \frac{125 - 122}{2.5}$$

$$z = \frac{3}{2.5} = \frac{30}{25} = 1.2$$

norm.cdf(z)

1 — norm.cdf(z)

18 / 19

A new page has been created.



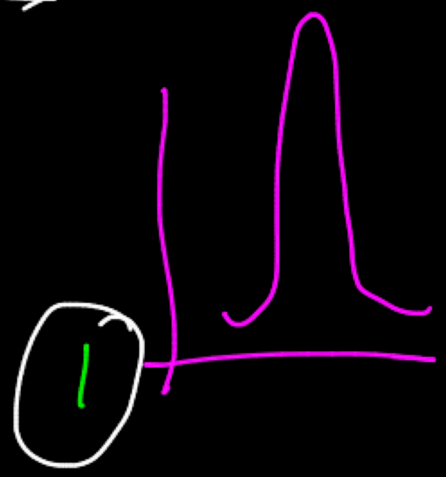
Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:01 PM

100 instead if ?

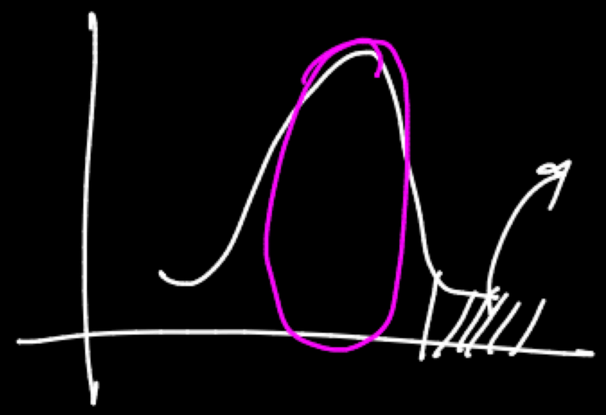
What is the probability of avg being greater than 125 mm?

$n = 100$ $\quad \sigma = 10$

✓ Std dev of sample mean $= \frac{\sigma}{\sqrt{n}} = \frac{10}{10} = 1$



✓ $z = \frac{125 - 122}{1} = 3$



1 − norm.cdf(z) ↵

0.0013 ⇒ 0.13

20 / 20

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:03 PM

⇒ In an e-commerce website. the avg purchase amount per customer is $80 with std-dev is of $15.

If we randomly select a sample of 50 customers, what is the probability that the avg purchase amount in the sample will be less than $75 ?

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:33 PM

=>

n > 30

2050

=>

Agriculture

scientist

fertilizer

50

8.5

wheat

1000 acers

15 acers

400 pounds population

400 pounds/g crop yield

Cursor selected

22 / 23

Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 10 Jul 11:15 PM

140 cr

sample

100 000

highs

Avg